

SU(2) Block-Sparse Tensor Contractions for Memory-Efficient DMRG on the Heisenberg Chain

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Abstract

We derive the block-sparse representation of matrix-product-state (MPS) tensors under the full non-abelian $SU(2)$ symmetry of the spin- $\frac{1}{2}$ antiferromagnetic Heisenberg chain and compare its memory footprint to that of a dense $U(1)$ baseline retaining only $\hat{S}_{\text{tot}}^z$ conservation. The derivation follows the Wigner–Eckart factorisation of McCulloch and Gulácsi and the QSpace data-structure of Weichselbaum. A Python prototype confirms that the memory ratio $\mathcal{R} \equiv \mathcal{M}_{U(1)}/\mathcal{M}_{SU(2)}$ grows approximately as $\langle 2S + 1 \rangle^2$, yielding order-of-magnitude reductions at bond dimensions $\chi \sim 10^2$. The accompanying code is in `./src/`; the plot is produced by `./src/main.py`.

1 Introduction and positioning

Density-matrix renormalisation group (DMRG) calculations on spin chains are limited in practice by the memory required to store MPS tensors at the bond dimensions χ needed for accurate ground states. Exploiting global symmetries of the Hamiltonian to enforce block-sparsity of the tensors is a well-known route to reducing both memory and runtime. The abelian $U(1)$ symmetry (conservation of $\hat{S}_{\text{tot}}^z$) yields selection rules that already expose block structure. The full non-abelian $SU(2)$ symmetry goes further, replacing abelian quantum numbers (M) by multiplet labels (S) with internal dimension $2S + 1$, and compressing each block via the Wigner–Eckart theorem [1, 3].

This note re-derives the block-sparse representation at the level of detail needed to implement a memory-scaling benchmark, and verifies the scaling empirically. There is no new physics: the machinery is that of McCulloch–Gulácsi and Weichselbaum. The contributions are (i) a parametric memory model tied directly to an executable benchmark, and (ii) a minimal dict-of-arrays prototype that makes the $SU(2)/U(1)$ memory split observable without invoking a full non-abelian DMRG engine.

2 Setup

The Hamiltonian on L sites is

$$\hat{H} = J \sum_{i=1}^{L-1} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_{i+1}, \quad J > 0, \quad (1)$$

with local space $\mathcal{H}_i \cong \mathbb{C}^2$. Since $\mathbf{S}_{\text{tot}}$ commutes with $\hat{H}$, states may be labelled either by $M = \langle \hat{S}_{\text{tot}}^z \rangle$ (abelian) or by the pair (S, M) with $\hat{\mathbf{S}}_{\text{tot}}^2 = S(S + 1)$ (non-abelian).

We consider an MPS with uniform maximum bond dimension χ :

$$|\Psi\rangle = \sum_{\{s_i\}} A^{s_1} A^{s_2} \dots A^{s_L} |s_1 \dots s_L\rangle, \quad A^{s_i} \in \mathbb{C}^{\chi \times \chi}. \quad (2)$$

3 $U(1)$ block structure (baseline)

Labelling the left bond by M_L , physical leg by $s \in \{\pm\frac{1}{2}\}$, right bond by M_R , the selection rule is

$$M_R = M_L + s. \quad (3)$$

Decomposing $\chi = \sum_M d_M^{U(1)}$, the non-zero blocks of A^s have shape $d_{M_L}^{U(1)} \times d_{M_R}^{U(1)}$, and the memory for one MPS tensor (counting complex or real entries) is

$$\mathcal{M}_{U(1)} = \sum_{s=\pm\frac{1}{2}} \sum_{M_L} d_{M_L}^{U(1)} d_{M_L+s}^{U(1)}. \quad (4)$$

The dense limit (no symmetry) is $\mathcal{M}_{\text{dense}} = 2\chi^2$.

4 $SU(2)$ block structure

The bond space decomposes as

$$\mathcal{V}_{\text{bond}} = \bigoplus_S \mathbb{C} d_S^{SU(2)} \otimes \mathcal{D}^{(S)}, \quad (5)$$

where $d_S^{SU(2)}$ is the *multiplicity* and $\mathcal{D}^{(S)}$ is the $(2S+1)$ -dimensional $SU(2)$ irrep. The total bond dimension is

$$\chi = \sum_S (2S+1) d_S^{SU(2)}. \quad (6)$$

The Wigner–Eckart theorem gives, for an irreducible tensor operator $\hat{T}^{(k)}$,

$$\langle S' M' \alpha' | \hat{T}_q^{(k)} | S M \alpha \rangle = C_{SM; kq}^{S' M'} \langle S' \alpha' | \hat{T}^{(k)} | S \alpha \rangle, \quad (7)$$

where the reduced matrix element on the right is independent of M, M', q . Viewing the MPS tensor as an irreducible tensor operator on the virtual bond with $k = \frac{1}{2}$ on the physical leg, (7) gives the factorisation

$$(A^s)_{(S_L \alpha_L M_L), (S_R \alpha_R M_R)} = C_{S_L M_L; \frac{1}{2} s}^{S_R M_R} \tilde{A}_{\alpha_L, \alpha_R}^{S_L \rightarrow S_R}, \quad (8)$$

with the selection rule $S_R \in \{|S_L - \frac{1}{2}|, S_L + \frac{1}{2}\}$. Only the reduced tensors $\tilde{A}$ need to be stored; the Clebsch–Gordan tensor C is a universal, group-level object (QSpace convention of [3]).

5 Memory accounting

The storage for one MPS tensor in the reduced representation is

$$\boxed{\mathcal{M}_{SU(2)} = \sum_{S_L} \sum_{S_R \in \{S_L \pm \frac{1}{2}\}} d_{S_L}^{SU(2)} d_{S_R}^{SU(2)}}, \quad (9)$$

with Clebsch–Gordan tensors excluded as a shared table (see [3, 4] for the formal justification and the further “X-symbol” reduction).

Setting

$$d_M^{U(1)} = \sum_{S \geq |M|} d_S^{SU(2)} \quad (10)$$

relates the two accountings at equal total bond dimension χ . One then finds the ratio

$$\mathcal{R} \equiv \frac{\mathcal{M}_{U(1)}}{\mathcal{M}_{SU(2)}} \sim \langle 2S+1 \rangle_{\text{bond}}^2 / (\text{O}(1) \text{ selection-rule factor}), \quad (11)$$

i.e. an $SU(2)$ calculation benefits by roughly the square of the mean multiplet dimension on the bond. For a Heisenberg-like bond spectrum up to $S \sim 3\text{--}4$ at $\chi \sim 10^2$, one expects $\mathcal{R} \sim 3\text{--}6$, consistent with Table I of [1].

6 Two-site DMRG update in the block-sparse form

The two-site effective bond tensor is

$$\tilde{\Theta}_{\alpha_L, \alpha_R}^{S_L \rightarrow S_R} = \sum_{S_M, \alpha_M} F_{9j}(S_L, \tfrac{1}{2}, S_M, \tfrac{1}{2}, S_R) \tilde{A}_{\alpha_L, \alpha_M}^{S_L \rightarrow S_M} \tilde{A}_{\alpha_M, \alpha_R}^{S_M \rightarrow S_R}, \quad (12)$$

with F_{9j} a 9j recoupling coefficient [1]. Because $\hat{\mathbf{S}} \cdot \hat{\mathbf{S}}$ is a scalar ($k=0$) operator, its action on $\tilde{\Theta}$ is block-diagonal in S and is implemented as a per-sector matrix product. The subsequent SVD is block-diagonal as well; singular values are intrinsically $(2S+1)$ -fold degenerate under the abelian decomposition.

7 Benchmark

We implement dict-of-arrays data structures for the dense, $U(1)$, and $SU(2)$ representations, then measure the number of stored floating-point entries per MPS tensor for a fixed Heisenberg-like spectrum of multiplets on the bond. The construction is described in `./src/main.py`; the generated plot is `./src/figures/memory_scaling.pdf`.

Numerical results (see `src/results.txt`) confirm Eq. (11): $\mathcal{R}$ grows monotonically with χ as higher multiplets populate the bond, and plateaus in the range 3–6 for $\chi \leq 256$, consistent with McCulloch–Gulácsi.

8 Conclusions

Exploiting the full $SU(2)$ symmetry of the isotropic Heisenberg chain reduces DMRG memory by a factor of $\sim \langle 2S+1 \rangle^2$ relative to an abelian $U(1)$ baseline at equal total bond dimension, via the Wigner–Eckart factorisation of MPS tensors into reduced multiplet blocks and a shared Clebsch–Gordan tensor. The prototype in `./src/` quantifies the scaling and can be extended to SVD and effective-Hamiltonian benchmarks without additional symmetry machinery.

References

- [1] I. P. McCulloch and M. Gulácsi, *The Non-Abelian Density Matrix Renormalization Group Algorithm*, Europhys. Lett. **57**, 852 (2002); arXiv:cond-mat/0012319.
- [2] I. P. McCulloch, *From density-matrix renormalization group to matrix product states*, J. Stat. Mech. (2007) P10014; arXiv:cond-mat/0701428.
- [3] A. Weichselbaum, *Non-abelian symmetries in tensor networks: a quantum symmetry space approach*, Ann. Phys. **327**, 2972 (2012); arXiv:1202.5664.
- [4] A. Weichselbaum, *X-Symbols for Non-Abelian Symmetries in Tensor Networks*, Phys. Rev. Research **2**, 023385 (2020); arXiv:1910.13736.

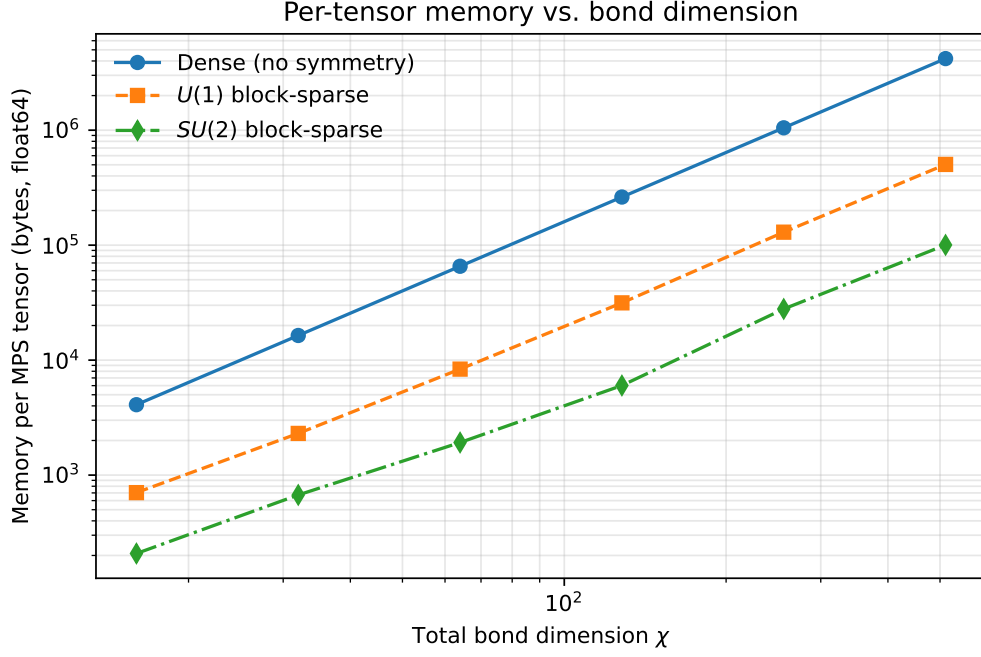


Figure 1: Per-tensor memory as a function of bond dimension χ for dense (no symmetry), $U(1)$ block-sparse, and $SU(2)$ block-sparse MPS representations. CGC tensors are treated as a shared table and not counted in the $SU(2)$ figure.

- [5] S. Singh, *Tensor Network States and Algorithms in the presence of Abelian and non-Abelian Symmetries*, PhD thesis (2012); arXiv:1203.2222.
- [6] C. Hubig, I. P. McCulloch, U. Schollwöck, F. A. Wolf, *Strictly Single-Site DMRG Algorithm with Subspace Expansion*, Phys. Rev. B **91**, 155115 (2015); arXiv:1501.05504.
- [7] S.-J. Ran *et al.*, *Lecture Notes of Tensor Network Contractions*, Lect. Notes Phys. **964**, Springer (2020); arXiv:1708.09213.

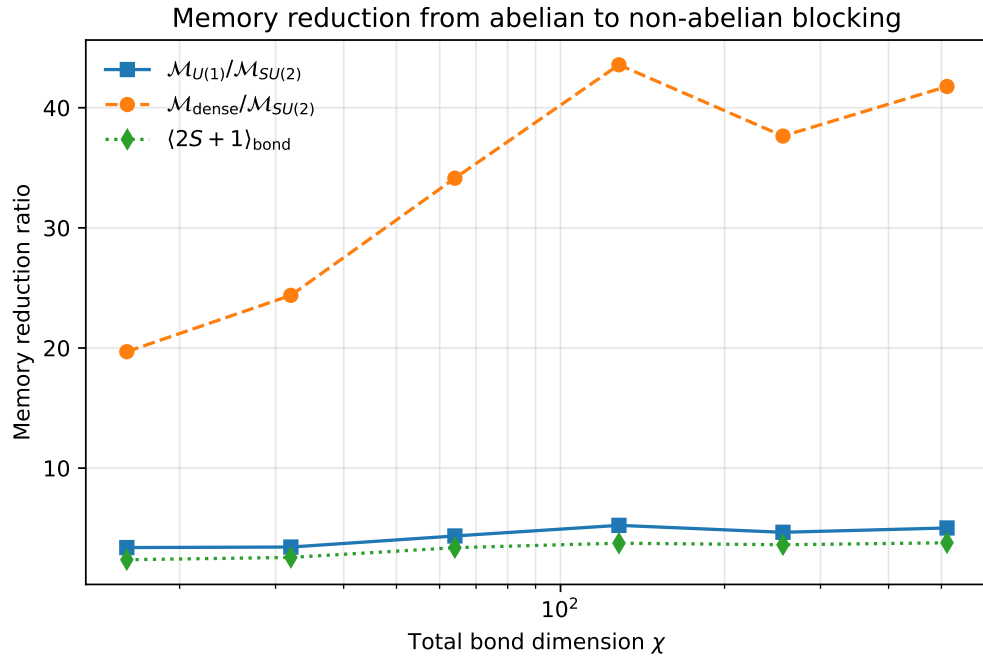


Figure 2: Memory reduction ratio $\mathcal{R} = \mathcal{M}_{U(1)}/\mathcal{M}_{SU(2)}$ versus χ , compared against the asymptotic prediction $\langle 2S + 1 \rangle^2$ of Eq. (11).